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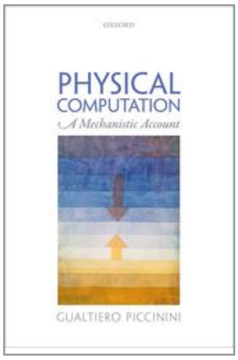

Physical Computation: A Mechanistic Account
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Information Processing

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Abstract and Keywords

This chapter distinguishes different notions of information and assesses the assimilation of computation to information processing. It explicates the notions of Shannon information, natural semantic information, and non-natural semantic information. The applications of Shannon information, which is non-semantic, are discussed. Broadly understood, semantic notions of information pertain to what a specific signal broadcast by an information source means. Natural semantic information and non-natural semantic information are distinguished, and each is considered in detail. Other notions of information are briefly discussed. The relations between computation and information processing are summarized in six theses. It is argued that while processing information (in any of these senses) entails computation, computation does not entail information processing.

Keywords: Shannon information, semantic information, natural semantic information, non-natural semantic information, computation, information processing

1. Computation = Information Processing?

Computation is typically used to process information, so much so that the notions of computation and information processing are often used interchangeably. This is built into some versions of the semantic account (Chapter 3), according to which there is no computation without representation. Here is a representative example: "I…describe the principles of operation of the human mind, considered as an *information-processing, or computational*, system" (Edelman 2008, 10, emphasis added). This statement presupposes that computation is the same as information processing. Is this right?

There are two immediate complications. First, as I argued beginning with Chapter 3, computation does not, in fact, require representation—computation can occur in the complete absence of representation. Second, there are several notions of both computations and information. As we have seen in previous chapters, computation comprises importantly different notions: computation in the generic sense, digital computation, analog computation, etc. The same is true of information. This chapter is devoted to distinguishing different notions of information and assessing the assimilation of computation to information processing.[1]

Information plays a central role in many disciplines. In the sciences of mind, information is invoked to explain cognition and behaviour (e.g., Miller 1951; Minsky 1968). In communication engineering, information is central to the design of efficient communication systems such as television, radio, telephone networks, and the internet (e.g., Shannon 1948; Fano 1961; Pierce 1980). A number of biologists have suggested explaining genetic inheritance in terms of the information carried by DNA sequences (e.g., Smith 2000; see also Godfrey-Smith 2000; Griffiths 2001). Animal communication theorists routinely characterize non-linguistic communication in terms of shared information (Bradbury and Vehrencamp 2000). Some philosophers maintain that information can provide a naturalistic grounding for the intentionality of mental states, namely, their being *about* states of affairs (Dretske 1981; Millikan **(p.226)** 2004). Finally, information plays important roles in several other disciplines, such as computer science, physics, and statistics.

To account for the different roles information plays in all these fields, more than one notion of information is required. I begin this chapter by distinguishing between three main notions of information: Shannon's non-semantic notion plus two notions of semantic information. After that, I'll go back to the relation between information processing and computation.

2. Shannon information

I use 'Shannon information' to designate the notion of information initially formalized by Claude Shannon (Shannon 1948).[2] I distinguish between information theory, which provides the formal definition of information, and communication theory, which applies the formal notion of information to engineering problems of communication. I begin by introducing two measures of information from Shannon's information theory. Later I will discuss how to apply them to the physical world.

Let X and Y be two discrete random variables taking values in $A_X=\{a_1,\ldots,a_n\}$ and $A_Y=\{b_1,\ldots,b_r\}$, respectively, with probabilities $p(a_1),\ldots,p(a_n)$ and $p(b_1),\ldots,p(b_r)$. We assume that $p(a_i)>0$ for all $i=1,\ldots,n$, $p(b_j)>0$ for all $j=1,\ldots,r$, $\sum_{i=1}^{n} p(a_i)=1$ and $\sum_{j=1}^{r} p(b_j)=1$.

Shannon's key measures of information are the following:

$$H(X) = -\sum_{i=1}^{n} p(a_i)\log_2 p(a_i)$$

$$I(X;Y) = \sum_{i=1,\,j=1}^{n,\,r} p(a_i,b_j)\log_2 \frac{p(a_i\mid b_j)}{p(a_i)}$$

H(X) is called 'entropy', because it's the same as the formula for measuring thermodynamic entropy; here, it measures the *average information* produced by the selection of values in $A_X=\{a_1,\ldots,a_n\}$.[3] We can think of H(X) as a weighted sum of n expressions of the form $I(a_i)=-\log_2 p(a_i)$. $I(a_i)$ is sometimes called the 'self-information' produced when X takes the value a_i. Shannon obtained the formula for *entropy* by setting a number of mathematical desiderata that any satisfactory measure of uncertainty **(p.227)** should satisfy, and showing that the desiderata could only be satisfied by the formula given above.[4]

Shannon information may be measured with different units. The most common unit is the "bit."[5] A bit is the information generated by a variable, such as X, taking a value a_i that has a 50 percent probability of occurring. Any outcome a_i with less than 50 percent probability will generate *more* than 1 bit; any outcome a_i with more than 50 percent probability will generate *less* than 1 bit. The choice of unit corresponds to the base of the logarithm in the definition of entropy. Thus, the information generated by X taking value a_i is equal to 1 bit when $I(a_i)=-\log_2 p_i=1$; that is, when $p_i=0.5$.[6]

Entropy has a number of important properties. First, it equals zero when X takes some value a_i with probability 1. This tells us that information as entropy presupposes uncertainty: if there is no uncertainty as to which value a variable X takes, the selection of that value generates no Shannon information.

Second, the closer the probabilities $p_1, \ldots, p_n$ are to having the same value, the higher the entropy. The more uncertainty there is as to which value will be selected, the more information is generated by the selection of a specific value.

Third, entropy is highest and equal to $\log_2 n$ when X takes every value with the same probability.

I(X;Y) is called 'mutual information'; it is the difference between the entropy characterizing X, on average, *before* and *after* Y takes values in A_Y. Shannon proved that X carries mutual information about Y whenever X and Y are statistically dependent, i.e., whenever it is not the case that $p(a_i, b_j)=p(a_i)p(b_j)$ for all i and j. This is to say that the transfer of mutual information between two sets of uncertain outcomes $A_X=\{a_1,\ldots,a_n\}$ and $A_Y=\{b_1,\ldots,b_r\}$ amounts to the *statistical dependency* between the occurrence of outcomes in A_X and A_Y. Information is *mutual* because statistical dependencies are symmetrical.

Shannon's measures of information have many applications. The best-known application is Shannon's own: communication theory. Shannon was looking for an optimal solution to what he called the "fundamental problem of communication" (Shannon 1948, 379), that is, the reproduction of messages from an information source to a destination. In Shannon's sense, any device that produces messages in a stochastic manner can count as an information source or destination.

(p.228)

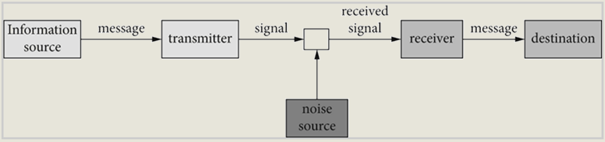

Figure 14.1 Shannon's depiction of a *communication system* (Shannon 1948).

Shannon distinguished three phases in the communication of messages: (a) the *encoding* of the messages produced by an *information source* into channel inputs, (b) the *transmission* of signals across a channel with random disturbances generated by a *noise source*, (c) the *decoding* of channel outputs back into the original messages at the *destination*. The process is summarized by the picture of a *communication system* shown in Figure 14.1.

Having distinguished the operations of *coding, transmission* and *decoding*, Shannon developed mathematical techniques to study how to perform them efficiently. Shannon's primary tool was the concept of *information*, as measured by entropy and mutual information.

A common misunderstanding of Shannon information is that it is a semantic notion. This misunderstanding is promoted by the fact that Shannon liberally referred to *messages* as carriers of Shannon information. But Shannon's notion of message is not the usual one. In the standard sense, a message has semantic content or meaning—there is something it stands for. By contrast, Shannon's messages need not have semantic content at all—they need not stand for anything.[7]

I will continue to discuss Shannon's theory using the term 'message,' but you should keep in mind that the usual commitments associated with the notion of message do not apply. The identity of a communication-theoretic message is fully described by two features: it is a physical structure distinguishable from a set of alternative physical structures, and it belongs to an exhaustive set of mutually exclusive physical structures selectable with well-defined probabilities.

Under these premises, to *communicate* a message produced at a source amounts to generating a second message at a destination that replicates the original message so as to satisfy a number of desiderata (accuracy, speed, cost, etc.). The sense in which the semantic aspects are irrelevant is precisely that messages carry information merely qua selectable, physically distinguishable structures associated with given probabilities. A seemingly paradoxical corollary of this notion of communication is that a **(p.229)** nonsense message such as “#r %h@” could in principle generate *more* Shannon information than a meaningful message such as “avocado”.

To understand why, consider an experiment described by a random variable X taking values a_1 and a_2 with probabilities $p(a_1) = 0.9999$ and $p(a_2) = 0.0001$ respectively. Before the experiment takes place, outcome a_1 is almost certain to occur, and outcome a_2 almost certain not to occur. The occurrence of both outcomes generates information, in the sense that it resolves the uncertainty characterizing the situation before the experiment takes place.

The occurrence of a_2 generates *more information* than the occurrence of a_1, because it is *less expectable*, or *more surprising*, in light of the prior probability distribution. This is reflected by the measure of self-information we introduced earlier: $I(a_i)=-\log_2 p(a_i)$. Given this measure, the higher $p(a_i)$ is, the lower $I(a_i)=-\log_2 p(a_i)$ is, and the lower $p(a_i)$ is, the higher $I(a_i)=-\log_2 p(a_i)$ is. Therefore, if “#r %h@” is a more surprising message than “avocado”, it carries more Shannon information, despite its meaninglessness.

Armed with his measures of non-semantic information, Shannon proved a number of seminal theorems that had a profound impact on the field of communication engineering. The “fundamental theorem for a noisy channel” established that it was theoretically possible to make the error rate in the transmission of information across a randomly disturbed channel as low as desired up until the point in which the source information rate in bits per unit of time becomes larger than the channel capacity, which is defined as the mutual information maximized over all input source probability distributions.[8] Notably, this is true regardless of the physical nature of the channel (e.g., true for electrical channels, wireless channels, water channels, etc.).

Communication theory has found many applications, most prominently in communication engineering, computer science, neuroscience, and psychology. For instance, Shannon information is commonly used by neuroscientists to measure the quantity of information carried by neural signals about a stimulus and estimate the efficiency of coding (what forms of neural responses are optimal for carrying information about stimuli) (Dayan and Abbott 2001, chap. 4; Baddeley et al. 2000). I now move on from Shannon information, which is non-semantic, to semantic information.

3. Semantic Information

Suppose that a certain process leads to the selection of signals. Think, for instance, of how you produce words when speaking. There are two dimensions of this process that are relevant for the purposes of information transmission. One is the uncertainty **(p.230)** that characterizes the speaker's word selection process as a whole—the probability that each word be selected. This is the non-semantic dimension that would interest a communication engineer and is captured by Shannon's theory. A second dimension of the word selection process concerns what the selection of each word *means* (in the context of a sentence). This would be the dimension of interest, for example, to a linguist trying to decipher a tribal language.

Broadly understood, semantic notions of information pertain to what a specific signal broadcast by an information source means. To address the semantics of a signal, it is neither necessary nor sufficient to know which other signals might have been selected instead and with what probabilities. Whereas the selection of, say, any of twenty-five equiprobable but distinct words will generate the same non-semantic information, the selection of each individual word will generate different semantic information depending on what that particular word means.

Semantic and non-semantic notions of information are both connected with the reduction of uncertainty. In the case of non-semantic information, the uncertainty has to do with which among many possible signals is selected. In the case of semantic information, the uncertainty has to do with which among many possible states of affairs is the case.

To tackle semantic information, let's begin with Grice's distinction between two kinds of meaning that are sometimes conflated, namely, natural and non-natural meaning (Grice 1957). Natural meaning is exemplified by a sentence such as 'those spots mean measles', which is true—Grice claimed—just in case the patient has measles. Non-natural meaning is exemplified by a sentence such as 'those three rings on the bell (of the bus) mean that the bus is full' (Grice 1957, 85), which is true even if the bus is not full.

I extend the distinction between Grice's two types of *meaning* to a distinction between two types of *semantic information*: *natural information* and *non-natural information*.[9] Spots carry natural information about measles because there is a reliable physical correlation between measles and spots. By contrast, the three rings on the bell of the bus carry non-natural information about the bus being full by virtue of a convention.[10] I will now consider each notion of semantic information in more detail.

3.1 Natural (Semantic) Information

When smoke carries information about fire, the basis for this informational link is the causal relation between fire and smoke. By the same token, when spots carry **(p.231)** information about measles, the basis for this informational link is the causal relation between measles and spots. Both are examples of natural semantic information.

The most basic task of an account of natural semantic information is to specify the relation that has to obtain between a source (e.g., fire) and a signal (e.g., smoke) for the signal to carry natural information about the source. Following Dretske (1981), I discuss natural information in terms of correlations between event types. On the view I propose, an event token *a* of type *A* carries natural information about an event token *b* of type *B* just in case *A reliably correlates* with *B*.

Reliable correlations are the sorts of correlations information users can count on to hold in some range of future and counterfactual circumstances. For instance, smoke-type events reliably correlate with fire-type events, spot-type events reliably correlate with measles-type events, and ringing-doorbell-type events reliably correlate with visitors-at-the-door-type events. It is by virtue of these correlations that we can dependably infer a fire-token from a smoke-token, a measles-token from a spots-token, and a visitors-token from a ringing-doorbell-token.

Yet correlations are rarely perfect. Smoke is occasionally produced by smoke machines, spots are occasionally produced by mumps, and doorbell rings are occasionally produced by naughty kids who immediately run away. Dretske (1981) and most subsequent theorists of natural information have disregarded imperfect correlations, either because they believe that they carry no natural information (Cohen and Meskin 2006) or because they believe that they do not carry the sort of natural information necessary for knowledge (Dretske 1981). Dretske focused only on cases in which signals and the events they are about are related by nomically underwritten perfect correlations.

This is unfortunate because, although not necessarily knowledge-causing, the transmission of natural information by means of imperfect yet reliable correlations is what underlies the central role natural information plays in the descriptive and explanatory efforts of many sciences. In other words, most of the natural information signals carry in real world environments is *probabilistic*: signals carry natural information to the effect that *o* is *probably G*, rather than natural information to the effect that, with nomic certainty, *o* is *G*. In the present account, all-or-nothing natural information—the natural information that *o* is *G* with certainty—is a special, limiting case of probabilistic natural information.

Unlike the traditional (all-or-nothing) notion of natural information, this probabilistic notion of natural information is applicable to the sorts of signals studied by many empirical sciences, including the science of cognition. Organisms survive and reproduce by tuning themselves to reliable but imperfect correlations between internal variables and environmental stimuli, as well as between environmental stimuli and threats and opportunities. In comes the sound of a predator, out comes running. In comes the redness of a ripe apple, out comes approaching.

(p.232) But organisms do make mistakes, after all. Some mistakes are due to the reception of probabilistic information about events that fail to obtain.[11] For instance, sometimes non-predators are mistaken for predators because they sound like predators; or predators are mistaken for non-predators because they look like non-predators.

Consider a paradigmatic example from ethology. Vervet monkeys' alarm calls appear to be qualitatively different for three classes of predators: leopards, eagles, and snakes (Struhsaker 1967; Seyfarth and Cheney 1992). As a result, vervets respond to eagle calls by hiding behind bushes, to leopard calls by climbing trees, and to snake calls by standing tall.

A theory of natural information with explanatory power should be able to make sense of the connection between the reception of different calls and different behaviors. To make sense of this connection, we need a notion of natural information according to which types of alarm call carry information about types of predators. But any theory of natural information that demands perfect correlations between event types will not do, because as a matter of empirical fact when the alarm calls are issued the relevant predators are not always present.

The present notion of probabilistic natural information can capture the vervet monkey case with ease, because it only demands that informative signals *change the probability* of what they are about. This is precisely what happens in the vervet alarm call case, in the sense that the probability that, say, an eagle/leopard/snake is present is significantly higher given the eagle/leopard/snake call than in its absence.

Notably, this notion of natural information is graded: signals can carry more or less natural information about a certain event. More precisely, the higher the difference between the probability that the eagle/leopard/snake is present given the call and the probability that it is present without the call, the higher the amount of natural information carried by an alarm call about the presence of an eagle/leopard/snake.

This approach to natural information allows us to give an information-based explanation of why vervets take refuge under bushes when they hear eagle calls (*mutatis mutandis* for leopard and snake calls). They do because they have received an ecologically significant amount of probabilistic information that an eagle is present.

To sum up, an event's failure to obtain is compatible with the reception of natural information about its obtaining, just like the claim that the probability that *o* is *G* is high is compatible with the claim that *o* is not *G*. No valid inference rules take us from claims about the transmission of probabilistic information to claims about how things turn out to be.[12]

(p.233) 3.2 Non-natural (Semantic) Information

Cognitive scientists routinely say that cognition involves the processing of information. Sometimes they mean that cognition involves the processing of natural information. At other times, they mean that cognition involves the processing of non-natural information. This second notion of information is best understood as the notion of *representation*, where a (descriptive) representation is by definition something that can get things wrong.[13]

Some cognitive scientists simply assimilate representation with *natural* semantic information, assuming in effect that what a signal represents is what it reliably correlates with. This is a weaker notion of representation than the one I endorse. Following the predominant usage in the philosophical literature, I reserve the term 'representation' for states that can get things wrong or misrepresent (e.g., Millikan 2004).

Bearers of natural information, I said, “mean” states of affairs in virtue of being physically connected to them. I provided a working account of the required connection: there must be a reliable correlation between a signal type and its source type. Whatever the right account may be, one thing is clear: in the absence of the appropriate physical connection, no natural information is carried.

Bearers of non-natural information, by contrast, need not be physically connected to what they are about in any direct way. Thus, there must be an alternative process by which bearers of non-natural information come to bear non-natural information about things they may not reliably correlate with. A convention, as in the case of the three rings on the bell of the bus, is a clear example of what may establish a non-natural informational link. Once the convention is established, error (misrepresentation) becomes possible.

The convention may either be explicitly stipulated, as in the rings case, or emerge spontaneously, as in the case of the non-natural information attached to words in natural languages. But non-natural information need not be based on convention (cf. Grice 1957). There may be other mechanisms, such as learning or biological evolution, through which non-natural informational links may be established. What matters for something to bear non-natural information is that, somehow, it stands for something else relative to a signal recipient.

An important implication of the present account is that semantic information of the non-natural variety does not entail truth. On the present account, false non- **(p.234)** natural information is a genuine kind of information, even though it is epistemically inferior to true information. The statement ‘water is not transparent’ gets things wrong with respect to the properties of water, but it does not fail to represent that water is not transparent. By the same token, the statement ‘water is not transparent’ contains false non-natural information to the effect that water is not transparent.

Most theorists of information have instead followed Dretske (Dretske 1981; Millikan 2004; Floridi 2005) in holding that false information, or misinformation, is not really information. This is because they draw a sharp distinction between information, understood along the lines of Grice's natural meaning, and representation, understood along the lines of Grice's non-natural meaning.

The reason for drawing this distinction is that they want to use natural information to provide a naturalistic reduction of representation (or intentionality). For instance, according to some teleosemantic theories the only kind of information carried by signals is of the natural variety, but signals come to *represent* what they have the function of carrying natural information about (e.g., Millikan 1984). According to theories of this sort, what accounts for how representations can get things wrong is the notion of biological function: representations get things wrong whenever they fail to fulfill their biological function.

But my present goal is not to naturalize intentionality. Rather, my goal is to understand the central role played by information and computation in computer science and cognitive science. If scientists used 'information' only to mean natural information, I would happily follow the tradition and speak of information exclusively in its natural sense. The problem is that the notion of information as used in the special sciences often presupposes representational content.

Instead of distinguishing sharply between information and meaning, I distinguish between natural information, understood roughly along the lines of Grice's natural meaning (cf. Section 3.1), and non-natural information, understood along the lines of Grice's non-natural meaning. We lose no conceptual distinction, while we gain an accurate characterization of how the concept of information is used in the sciences. This is a good bargain.

We are now in a position to refine the sense in which Shannon information is non-semantic. Shannon information is non-semantic in the sense that Shannon's measures of information are not measures of semantic content or meaning, whether natural or non-natural. Shannon information is also non-semantic in the sense that, as Shannon himself emphasized, the signals studied by communication theory need not carry any non-natural information.

But the signals studied by communication theory always carry natural information about their source when there is mutual information in Shannon's sense between source and receiver. When a signal carries Shannon information, there is by definition a reliable correlation between source events and receiver events. For any receiver event, there is a nonzero probability of the corresponding source event. And as I defined natural information, *A* carries natural information about *B* just in **(p.235)** case *A* reliably correlates with *B*—thus, receiver events carry natural information about source events whenever there is mutual information between the two. In addition, in virtually all practical applications of communication theory (as opposed to information theory more generally), the probabilities connecting source and receiver events are underwritten by a causal process.

For instance, ordinary communication channels such as cable and satellite television rely on signals that causally propagate from sources to receivers. Because of this, the signals studied by communication theory, in addition to producing a certain amount of mutual information in Shannon's sense based on the probability of their selection, also carry a certain amount of natural semantic information about their source (depending on how reliable their correlation with the source is). Of course, in many cases, signals also carry non-natural information about things other than their sources—but that is a contingent matter, to be determined case by case.

4. Other Notions of Information

There are other notions of information, such as *Fisher information* (Fisher 1935) and information in the sense of *algorithmic information theory* (Li and Vitányi 1997). There is even an all-encompassing notion of physical information. Physical information may be generalized to the point that every state of a physical system is defined as an information-bearing state. Given this all-encompassing notion, allegedly the physical world is, at its most fundamental level, constituted by information (Wolfram 2002; Lloyd 2006). If computation is defined as information processing in this sense, then this is just a variant formulation of ontic pancomputationalism, which I rejected in Chapter 4, Section 4.

In one or more of these senses of the term, computing entails processing information (cf. Milkowski 2013, 42–51). But these notions of information are not directly relevant to our present concerns, so I set them aside.

Nir Fresco and Marty Wolf have recently introduced an interesting notion of *instructional information* (Fresco 2013, Chapter 6; Fresco and Wolf 2014). Fresco and Wolf's formulation relies on notions of data and meaning that I don't fully understand, so I'll offer an alternative account that captures the gist of what they are after. Let *s* be a control signal affecting system *C*:

> *s* carries *instructional information e* within system *C* just in case when *C* acts in accordance with *s, C* generates action *e*.

Notice that *s* may be but *doesn't have to be* a string of digits that acts as an instruction in the sense of Chapter 9. Signal *s* is *any* signal external to *C* that activates a capacity of *C*. For example, consider a Boolean circuit that can perform two different operations depending on a single control digit. That single control digit counts as carrying instructional information. In any case, acting in accordance with *s* means doing what *s* causes within the system (i.e., action *e*), which may be seen as what *s* **(p.236)** says to do. Thus, Fresco and Wolf's notion of instructional information gives rise to a more inclusive notion of instruction than the one discussed in Chapter 9. Fresco and Wolf maintain that "nontrivial" digital computation is the processing of discrete data in accordance with instructions in their sense. Since discrete data in Fresco and Wolf's sense are strings of digits and Fresco and Wolf's instructions are a kind of rule, Fresco and Wolf's account of "nontrivial" digital computation appears to be a digital version of the causal account of computation (Chapter 2) restricted to systems that respond to control signals. If we formulate Fresco and Wolf's account in terms of functional mechanisms (Chapter 6), then their account is equivalent to the mechanistic account of *digital* computations that respond to control signals (Chapter 7).

In what follows, I focus on the relations between computation and the following three notions of information: Shannon information (information according to Shannon's theory), natural information (truth-entailing semantic information based on a physical connection between signal and source), and non-natural information (non-truth-entailing semantic information).

The differences between these three notions can be exemplified as follows. Consider an utterance of the sentence ‘I have a toothache’. It carries Shannon information just in case the production of the utterance is a stochastic process that generates words with certain probabilities. The same utterance carries natural information about having a toothache just in case utterances of that type reliably correlate with toothaches. Carrying natural information about having a toothache entails that a toothache is more likely given the signal than in the absence of the signal. Finally, the same utterance carries non-natural information just in case the utterance has non-natural meaning in a natural or artificial language. Carrying non-natural information about having a toothache need not raise the probability of having a toothache.

5. How Computation and Information Processing Fit Together

Do the vehicles of computation necessarily bear information? Is information processing necessarily carried out by means of computation? What notion of information is relevant to the claim that a computational system is an information processor? Answering these questions requires combining computation and information in several of the senses I have discussed.

I will focus on two main notions of computation, digital and generic, and three main notions of information processing: processing Shannon information, processing natural semantic information and processing non-natural semantic information. It’s time to see how they fit together.

I will argue that the vehicles over which computations are performed may or may not carry information. Yet, as a matter of contingent fact, the vehicles over which **(p.237)** most computations are performed generally do carry information, in several senses of the term. I will also argue that information processing must be carried out by means of computation in the generic sense, although it need not be carried out by computation in any more specific sense.

Before I begin, I should emphasize that there is a trivial sense in which computation entails the processing of Shannon information. I mention it now but will immediately set it aside because it is not especially relevant to computer science and cognitive science.

Consider an electronic digital computer. Each of its memory cells is designed to stabilize on one of two thermodynamically macroscopic states, called '0' and '1'. Suppose that, as soon as the computer is switched on, each memory cell randomly goes into one of those states. A nonzero Shannon entropy is associated with this state initialization process, since it is a random process, and manipulation of randomly generated initial states by the computer amounts to the processing of Shannon information in this trivial sense.

Note, however, that the digits created by random initialization of the memory cells carry no non-natural semantic information, since the digits in question do not have any conventional semantic content. They carry no functionally relevant natural semantic information either, since they do not reliably correlate with any source other than the thermal component of the electrons' dynamics in the memory cell at power up. Random noise is not the kind of source of natural information that is relevant to the way computers are used or to the explanation of cognition. Thus, the digits do not carry any natural semantic information beyond the theoretically unimportant natural information they carry about what caused them—whatever that may be.

Since a computer manipulates its digits and, in the above sense, a nonzero Shannon information is associated with randomly generated digits, it follows that a computer processing randomly generated digits is processing information-bearing states. In this trivial sense, such a computer is necessarily a processor of Shannon information. But here I am not interested in whether computation can be regarded as processing information in this trivial sense. I am interested in how computation fits with the processing of Shannon information, natural semantic information, and non-natural semantic information *about sources in the system's distal environment*—the sources the system needs to respond to. I will take up each of these in turn.

5.1 Computation and the Processing of Shannon Information

The notion of *processing* Shannon information is not entirely clear. Shannon information is not a property of individual signals, which can be manipulated as the signal is manipulated. Instead, Shannon entropy and mutual information quantify statistical properties of a selection process and a communication channel as a whole.

The first thing to note is that, strictly speaking, computational vehicles need not *carry* Shannon information at all. This is because Shannon information **(p.238)** requires random variables—deterministic variables carry no Shannon information whatsoever—whereas computation is compatible with deterministic variables. For instance, consider again a digital computer. For its inputs to be regarded as carriers of Shannon information, they have to be selected in a stochastic way. But the computer works just the same whether its inputs are selected deterministically or probabilistically. The same point can be made about other computing devices, such as analog computers and neural networks. Thus, computation does not entail the processing of Shannon information.

Another relevant observation is that computation need not *create* Shannon information. This is because computation may be either probabilistic or deterministic—in most real-world applications, computation is deterministic. Deterministic computation cannot create Shannon information—it can only conserve or destroy it. More precisely, Shannon information is conserved in logically reversible computations (which are necessarily deterministic), is reduced in logically irreversible deterministic computations, and may be created, conserved, or destroyed in noisy computations.

These points should not obscure the fundamental importance of communication theory to the understanding of computing systems. Even though, strictly speaking, computing systems need not process Shannon information, it is useful to assume that they do. That is, it is useful to characterize the inputs, internal states, and outputs of computing systems and their components stochastically and analyze system performance and resource requirements using information-theoretic measures and techniques. On this view, the components of computing systems may be treated as sources and receivers in the sense of communication theory (Winograd and Cowan 1963). This allows engineers to design efficient computer codes and effective communication channels within computing systems. Similar techniques may be used to analyze neural codes and signal transmission within the nervous system.

So far, I have discussed whether computation is the processing of Shannon information. What about the converse? Processing Shannon information may or may not be done by means of digital computation. First, for Shannon information processing to be done by digital computing, the information must be produced and carried by strings of digits. But Shannon information can also be produced and carried by continuous signals, which may be processed by analog means. Second, even when the bearers of Shannon information are digits, there exist forms of processing of such digits other than digital computation. Just to give an example, a digital-to-analog converter transforms digits into analog signals.

Shannon information processing is, however, a form of computation in the generic sense. As I defined it, generic computation is the functional manipulation of any medium-independent vehicle. Shannon information is a medium-independent notion, in the sense that whether Shannon information can be associated with a given vehicle does not depend on its specific physical properties, but rather on its probability of occurrence relative to its physically distinguishable alternatives. Thus, generic computation is broad enough to encompass the processing of Shannon **(p. 239)** information. In other words, if a vehicle carries Shannon information, its processing is a computation in the generic sense.

The bottom line is this: although, strictly speaking, computation need not process vehicles that carry Shannon information, it usually does and, at any rate, it is useful to assume that it does. Conversely, the processing of Shannon information amounts to some form of generic computation. Communication theory places constraints on any kind of signal transmission (and hence on the processing of any information carried by the transmitted signals) within computing systems, including cognitive ones. This does not yet tell us what we are most interested in: how does computation relate to semantic information?

5.2 Computation and the Processing of Natural Information

Many people use ‘computation’ and ‘information processing’ interchangeably. What they often mean by ‘information processing’ is the processing of natural semantic information carried by the computation vehicles. This use of ‘information processing’ is common in the behavioral and brain sciences. It is thus important to examine whether and in what sense computation is the processing of natural information.

The notion of digital computation does not require that the computation vehicles carry natural information—or more precisely, digits are not required to carry natural information about the computing system’s *distal* environment.[14] In this section, I will focus on whether digital computation entails the processing of natural information about the system’s distal environment.

Granted, natural information is virtually ubiquitous—it is easy enough to find reliable correlations between physical variables. This being the case, the digits within a digital computer may well carry natural information about cognitively relevant sources, such as environmental stimuli that the computer responds to. Consider a car’s computer, which receives and responds to natural information about the state of the car. The computer uses feedback from the car to regulate fuel injection, ignition timing, speed, etc. In such a case, a digital computation will be a case of natural information processing.

But digits—the vehicles of digital computation—need not correlate reliably with anything (in the distal environment) in order for a digital computation to be performed over them. Thus, a digital computation may or may not constitute the processing of natural information (about the distal environment). If the computation vehicles *do* carry natural information (about the distal environment), this may be quite important. In our example, that certain digits carry natural information about **(p.240)** the state of the car explains why the car’s computer can regulate, say, fuel injection successfully. My point is simply that digital computation does not entail the processing of natural information about the distal environment.

This point is largely independent of the distinction between semantically interpreted and non-semantically-interpreted digital computation. A semantically interpreted digital computation is generally defined over the *non*-natural information carried by the digits—independently of whether they carry natural information (about the distal environment) or what specific natural information they carry. For instance, ordinary mathematical calculations carried out on a computer are defined over numbers, regardless of what the digits being manipulated by the computer reliably correlate with. Furthermore, it is possible for a digital computation to yield an incorrect output, which misrepresents the outcome of the operation performed. As we have seen, natural information cannot misrepresent. Thus, even semantically interpreted digital computation does not entail the processing of natural information (about the distal environment). (More on the relation between natural and non-natural information in the next sections.)

This point generalizes to generic computation, which includes, in addition to digital computation, at least analog computation. Nothing in the notion of analog computation mandates that the vehicles being manipulated carry natural information (about the distal environment). More generally, nothing in the notion of manipulating medium-independent vehicles mandates that such vehicles carry natural information (about the distal environment). Thus, computation may or may not be the processing of natural information.

What about the converse claim: that the processing of natural information must be carried out by means of computation? The answer is that yes, it must, although the computation need not be digital or of any specific kind. Natural information may be encoded by continuous variables, which cannot be processed by digital computers unless they are first encoded into strings of digits. Or it may be encoded by strings of digits, whose processing consists in converting them to analog signals. In both of these cases, natural information is processed, although not by means of digital computation.

But if natural information is processed, some kind or other of generic computation must be taking place. Natural information, like Shannon information, is a medium-independent notion. As long as a variable reliably correlates with another variable, it carries natural information about it. Any other physical properties of the variables are immaterial to whether they carry natural information. Since generic computation is just the processing of medium-independent vehicles, any processing of natural information amounts to a computation in the generic sense.

In sum, computation vehicles need not carry natural information, but natural information processing is necessarily some kind of generic computation—though not necessarily either digital or analog computation.

(p.241) 5.3 Computation and the Processing of Non-natural Information

Non-natural information is a central notion in our discourse about minds and computers. We attribute representational contents to each other's minds. We do the same with the digits manipulated by computers. I will now examine whether and in what sense computation is the processing of non-natural information.

Digital computation may or may not be the processing of non-natural information, and vice versa. This case is analogous to that of natural information, with one difference. In the case of *semantically interpreted* digital computation—digital computation that has semantic content by definition—most notions of semantic content turn digital computation into non-natural information processing. (The primary exception are notions of semantic content based solely on natural information; but since natural information is not representational, such notions would be inadequate to capture the main notion of semantic content—representational semantic content.) As a matter of fact, almost all (digital) computing conducted in artifacts *is* information processing, at least in the ordinary sense of *non-natural* information.

But the main question I am trying to answer is whether *necessarily*, digital computation is the processing of non-natural information. It *is* if we define a notion of digital computation that entails that the digits are representations. But as I have repeatedly pointed out, nothing in the notion of digital computation used by computer scientists and computability theorists mandates such a definition. The theoretically most important sense of digital computation is the one implicitly defined by the practices of computability theory and computer science, and which provides a mechanistic explanation for the behavior of digital computers. In this theoretically most important sense of digital computation, digital computation need not be the processing of non-natural information.

Conversely, the processing of non-natural information need not be carried out by means of digital computation. Whether it is depends on whether the non-natural information is digitally encoded and how it is processed. It may be encoded either digitally, by continuous variables, or by other vehicles. Even when encoded digitally, non-natural information may be manipulated by digital computation or by other processes, as in digital-to-analog conversion.

Generic computation may or may not constitute the processing of non-natural information, for similar reasons. Nothing in the notion of analog computation mandates that the vehicles being manipulated carry non-natural information. They could be entirely meaningless and the computations would proceed just the same. More generally, nothing in the notion of manipulating medium-independent vehicles mandates that such vehicles carry non-natural information.

As in the case of natural information, the converse does not hold. Instead, non-natural semantic information processing is necessarily done by means of computation in the generic sense. For non-natural semantic information is medium-independent—it's defined in terms of the association between an information-carrying **(p.242)** vehicle and what it stands for, without reference to any physical properties of the vehicles. Since generic computation is defined as the processing of medium-independent vehicles, the processing of non-natural semantic information (i.e., the processing of vehicles carrying non-natural semantic information) is computation in the generic sense.

Our work is not done yet. I defined non-natural information in terms of the association between bearers of non-natural information and what they stand for. This representation relation requires explication.

In recent decades, philosophers have devoted a large literature to representation. They have articulated several theoretical options. One option is anti-naturalism: representation cannot and need not be explained or reduced in naturalistic or mechanistic terms—the representational character of cognition simply shows that psychology and perhaps neuroscience are *sui generis* sciences. A second option is anti-realism: representation is not a real (i.e., causally efficacious and explanatory) feature of cognition, although ascriptions of non-natural information-bearing states may be heuristically useful (Egan 2010). Finally, naturalists look for explanations of representation in non-intentional terms (Rupert 2008).

A prominent family of naturalistic theories attempts to reduce non-natural information to natural information—plus, perhaps, some other naturalistic ingredients. As I mentioned earlier, several efforts have been made to this effect (Dretske 1981; Millikan 2004; Dretske 1988; Barwise and Seligman 1997; Fodor 1990). According to these theories, non-natural information is, at least in part, natural information. But theories according to which non-natural information reduces to natural information are only one family of naturalistic theories among others. According to other naturalistic theories, non-natural information reduces to things other than natural information (e.g., Millikan 1984; Harman 1987; Papineau 1987; Grush 2004; Ryder 2004).

Summing up the contents of this section, I have reached two main conclusions. First, computation may or may not be the processing of non-natural information—with the exception of semantically interpreted computation, which is *not* the core notion of computation. Second, the processing of non-natural information is a kind of generic computation.

6. Computation is Not Information Processing

In this chapter, I have charted some conceptual relations between computation and information processing. ‘Computation’ and ‘information processing’ are commonly used interchangeably, which presupposes that they are roughly synonymous terms. If that were true, computation would entail information processing, and information processing would entail computation. Alas, things are not so simple. Each of the two assimilated notions has more than one meaning, and the entailment relations do **(p.243)** not hold on many legitimate ways of understanding computation and information processing.

Here are my main conclusions:

Question: Is digital computation the same as information processing?
Thesis 1: Digital Computation vs. Processing Shannon Information
(a) Digital computation does not entail the processing of Shannon information;
(b) The processing of Shannon information does not entail digital computation.

Thesis 2: Digital Computation vs. Processing Natural Information
(a) Digital computation does not entail the processing of natural information (about the distal environment);
(b) The processing of natural information does not entail digital computation.

Thesis 3: Digital Computation vs. Processing Non-natural Information
(a) Non-semantically interpreted digital computation does not entail the processing of non-natural information;
(b) Semantically interpreted digital computation entails the processing of non-natural information;
(c) The processing of non-natural information does not entail digital computation.

Question: Is generic computation the same as information processing?
Thesis 4: Generic Computation vs. Processing Shannon Information

(a) Generic computation does not entail the processing of Shannon information;
(b) The processing of Shannon information entails generic computation.

Thesis 5: Generic Computation vs. Processing Natural Information

(a) Generic computation does not entail the processing of natural information (about the distal environment);
(b) The processing of natural information entails (semantically interpreted) generic computation.

Thesis 6: Generic Computation vs. Processing Non-natural Information

(a) Non-semantically interpreted generic computation does not entail the processing of non-natural information;
(b) Semantically interpreted generic computation entails the processing of non-natural information;
(c) The processing of non-natural information entails (semantically interpreted) generic computation.

These six theses summarize some of the relations of conceptual entailment, or lack thereof, between ‘computation’ and ‘information processing’, depending on what each notion is taken to mean.

Having dealt with the relation between computation and information processing, the final two chapters will address another pervasive and controversial topic: the computational power of physical systems.

Notes:

([1]) This chapter derives mostly from Piccinini and Scarantino 2011, so Andrea Scarantino deserves partial credit for most of what is correct here.

([2]) Shannon was building on important work by Boltzmann, Szilard, Hartley, and others (Pierce 1980).

([3]) The logarithm to the base b of a variable x—expressed as log_b x—is defined as the power to which b must be raised to get x. In other words, $\log_b x=y$ if and only if $b^y=x$. The expression $0 \log_b 0$ in any of the addenda of H(X) is stipulated to be equal to 0. Shannon (1948, 379) pointed out that in choosing a logarithmic function he was following Hartley (Hartley 1928) and added that logarithmic functions have nice mathematical properties, are more useful practically because a number of engineering parameters “vary linearly with the logarithm of the number of possibilities,” and are “nearer to our intuitive feeling as to the proper measure” of information.

([4]) The three mathematical desiderata are the following: (i) The entropy H should be continuous in the probabilities p_i, (ii) The entropy H should be a monotonic increasing function of n when $p_i=1/n$, and (iii) If $n=b_1+\ldots+b_k$ with b_i positive integer, then $H(1/n,\ldots,1/n)=H(b_1/n,\ldots,b_k/n)+\sum i=1k\, b_i/n\, H(1/b_i,\ldots,1/b_i)$. Shannon further supported his interpretation of H as the proper measure of information by demonstrating that the channel capacity required for most efficient coding is determined by the entropy (Shannon 1948; see Theorem 9 in Section 9).

([5]) Shannon credits John Tukey, a computer scientist at Bell Telephone Laboratories, with introducing the term in a 1947 working paper.

([6]) $-\log_2 0.5=1$.

([7]) Furthermore, Shannon’s messages don’t even have to be strings of digits of finitely many types; on the contrary, they may be continuous variables. I defined entropy and mutual information for discrete variables, but the definitions can be modified to suit continuous variables (Shannon 1948, Part III).

([8]) Shannon defined the channel capacity C as follows: $M\,a\,x\,P\,(\,a\,)\,I\,(\,X;Y\,) = M\,a\,x\,P\,(\,a\,)\,[\,H(X)\text{-}H(X|Y)\,]$. The conditional entropy is calculated as follows: $H(\,X|Y\,) = \sum i = 1\,,\,j = 1\,n\,,\,r\,p\,(\,a\,i\,,\,b\,j\,)\,\log 2\;1\;p\,(\,a\,i\,|\,b\,j\,)$.

([9]) By calling this kind of information *non-natural*, I am not taking a stance on whether it can be naturalized, that is, reduced to some more fundamental natural process. I am simply using Grice's terminology to distinguish between two importantly different notions of semantic information.

([10]) This is not to say that conventions are the only possible source of non-natural meaning. For further discussion, see (Grice, 1957).

([11]) This is not to say that natural information is enough to explain why acting on the basis of received natural information sometimes constitutes a mistake and sometimes it does not.

([12]) A consequence of this point is that the transmission of natural information entails nothing more than the truth of a probabilistic claim (Scarantino and Piccinini 2010). It follows that the present distinction between natural and non-natural meaning/information differs from Grice's original distinction in one important respect. On the present view, there is nothing objectionable in holding that *those spots carry natural information about measles, but he doesn't have measles*, provided measles are more likely given those spots than in the absence of those spots.

([13]) There are also imperative representations, such as desires. And there are representations that combine descriptive and imperative functions, such as honeybee dances and rabbit thumps (cf. Millikan 2004). For simplicity, I focus on descriptive representations.

([14]) At least for deterministic computation, there is always a reliable causal correlation between later states and earlier states of the system as well as between initial states and whatever caused those initial states—even if what caused them are just the thermal properties of the immediate surroundings. In this sense, (deterministic) digital computation entails natural information processing. But the natural information that is always "processed" is not about the distal environment, which is what theorists of cognition are generally interested in.

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